If three years of up to 25 per cent unemployment could turn a seemingly prospering democracy into the most brutal totalitarian regime in history, what might happen to democracies when automation causes even bigger upheavals in the job market of the twenty-first century? Nobody knows what the job market will look like in 2050, or even in 2030, except that it will look very different from today. AI and robotics will change numerous professions, from harvesting crops to trading stocks to teaching yoga. Many jobs that people do today will be taken over, partly or wholly, by robots and computers.

Of course, as old jobs disappear, new jobs will emerge. Fears of automation leading to large-scale unemployment go back centuries, and so far they have never materialised. The Industrial Revolution put millions of farmers out of agricultural jobs and provided them with new jobs in factories. It then automated factories and created lots of service jobs. Today many people have jobs that were unimaginable thirty years ago, such as bloggers, drone operators and designers of virtual worlds. It is highly unlikely that by 2050 all human jobs will disappear. Rather, the real problem is the turmoil of adapting to new jobs and conditions. To cushion the blow, we need to prepare in advance. In particular, we need to equip younger generations with skills that will be relevant to the job market of 2050.

Unfortunately, nobody is certain what skills we should teach children in school and students in university, because we cannot predict which jobs and tasks will disappear and which ones will emerge. The dynamics of the job market may contradict many of our intuitions. Some skills that we have cherished for centuries as unique human abilities may be automated rather easily. Other skills that we tend to look down on may be far more difficult to automate.

For example, intellectuals tend to appreciate intellectual skills more than motor and social skills. But actually, it is easier to automate chess playing than, say, dishwashing. Until the 1990s, chess was often hailed as one of the prime achievements of the human intellect. In his influential 1972 book *What Computers Can't Do*, the philosopher Hubert Dreyfus studied various attempts to teach computers chess and noted that despite all these efforts computers were still unable to defeat even novice human players. This was a crucial example for Dreyfus's argument that computer intelligence is inherently limited.[9] In contrast, nobody thought that dishwashing was